Regression Performance Evaluation Metrics

Regression Performance Evaluation

Regression evaluation metrics measure how close the predicted values are to the actual values.

1. Mean Absolute Error (MAE)

Meaning:

MAE calculates the average absolute difference between actual and predicted values.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|Y_i - \hat{Y}_i|$$

Where,

- Y_i = Actual value
- $\hat{Y}_i$ = Predicted value
- n = Number of observations

Interpretation:

Lower MAE indicates better prediction accuracy.

2. Mean Squared Error (MSE)

Meaning:

MSE calculates the average squared difference between actual and predicted values.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

Interpretation:

Larger errors are penalized more because errors are squared.

3. Root Mean Squared Error (RMSE)

Meaning:

RMSE is the square root of MSE and represents error in the original unit.

$$RMSE = \sqrt{MSE}$$

Interpretation:

RMSE is easier to interpret because it has the same unit as the target variable.

4. Mean Absolute Percentage Error (MAPE)

Meaning:

MAPE represents the average prediction error as a percentage.

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{Y_i - \hat{Y}_i}{Y_i}\right| \times 100$$

Interpretation:

MAPE tells how much the prediction differs from the actual value in percentage terms.

5. R^2 Score (Coefficient of Determination)

Meaning:

R^2 measures how much variation in the output variable is explained by the regression model.

$$R^2 = 1 - \frac{\sum(Y_i - \hat{Y}_i)^2}{\sum(Y_i - \bar{Y})^2}$$

Where,

- Y_i = Actual value
- $\hat{Y}_i$ = Predicted value
- $\bar{Y}$ = Mean of actual values

Interpretation:

$$R^2 = 0$$

Model explains no variation.

$$R^2 = 1$$

Model explains 100% variation.

$$R^2 = 0.92$$

The model explains 92% of the variation in the data.

Summary Table

Metric	Purpose
MAE	Average absolute prediction error
MSE	Average squared prediction error
$RMSE$	Error in original unit
$MAPE$	Average percentage error
R^2	Variation explained by model